KEY TO SYMBOLS AND TERMS

Soil and rock types


TOP – TOPSOIL


SM – SAND, silty


CH – CLAY, high plastic


SP – SAND, poorly graded


Till overlay: scattered clasts on the soil pattern


CI – CLAY, medium plastic


FILL – FILL


ML – SILT, low plastic

Standpipe installation


Concrete


Cement-bentonite grout


Bentonite seal


Silica sand pack

Consistency (cohesive) · Compactness (cohesionless) · Proportions

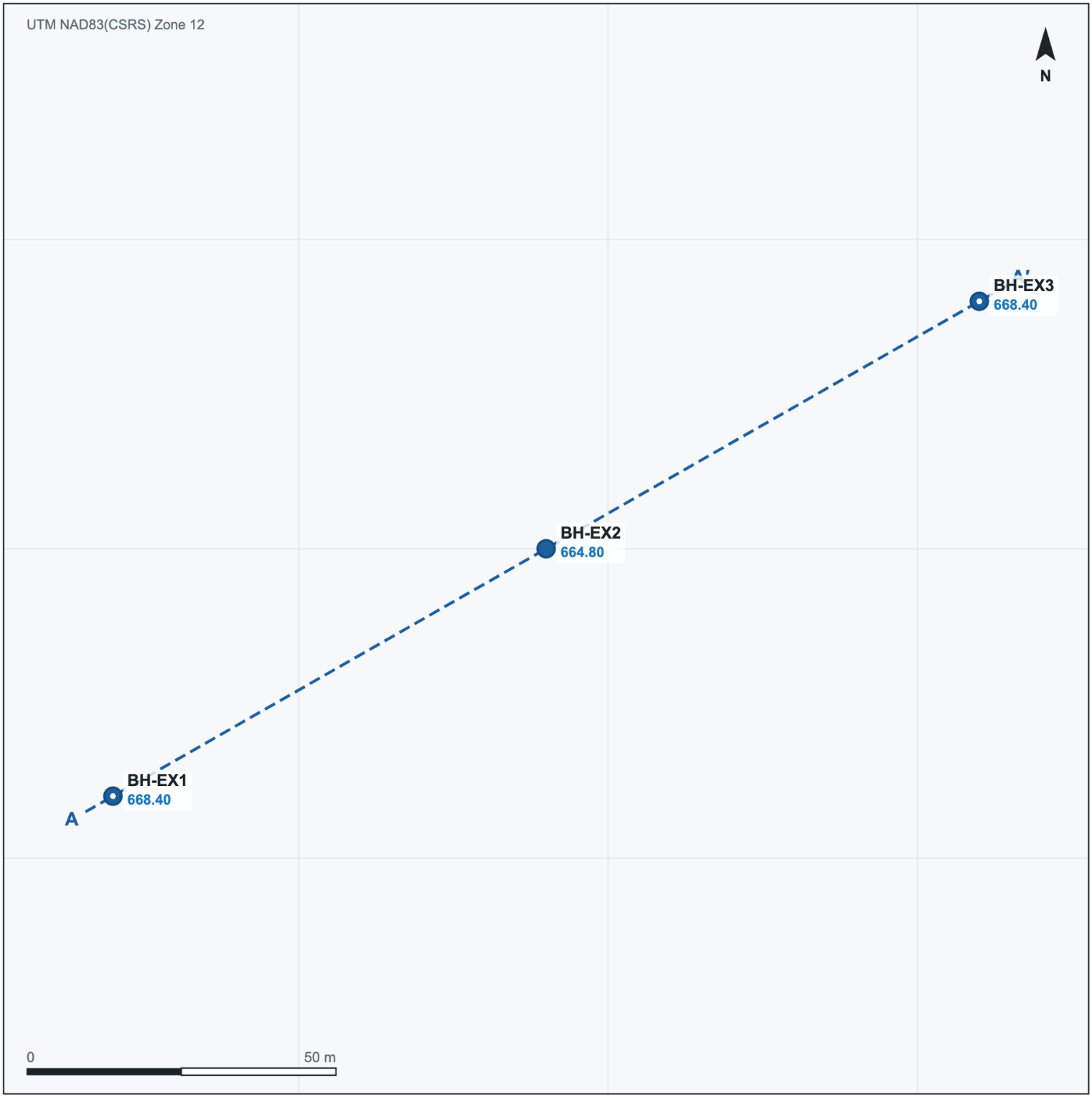
Very soft	< 12 kPa	**Very loose**	N 0–4	**trace**	< 10 %
Soft	12–25	**Loose**	4–10	**some**	10–20 %
Firm	25–50	**Compact**	10–30	**-y / -ey**	20–35 %
Stiff	50–100	**Dense**	30–50	**and**	> 35 %
Very stiff	100–200	**Very dense**	> 50		
Hard	> 200				

Samples, tests and symbols

SS	Split spoon (SPT, ASTM D1586)	**ST**	Shelby tube
G / A / B	Grab / auger / bulk	**C**	Rock core (TCR, RQD)
N	SPT blows, blows/0.3 m, field	**N60, (N1)60**	Energy / overburden corrected
w, PL, LL	Moisture content, Atterberg limits (%)	**PP, Su**	Pocket penetrometer, field vane (kPa)
SO4	Water-soluble sulphate (%), CSA A23.1	**▽ / ▼**	Seepage while drilling / water level
S	Slough upon completion		

Notes

Refer to the geotechnical report for interpretation. Subsurface conditions are shown at the borehole location only and may vary between boreholes. Read this log together with the accompanying report.


UTM NAD83(CSRS) Zone 12
N
BH-EX3
668.40
BH-EX2
664.80
BH-EX1
668.40
A
A'
0
50 m

XTES Environmental Ltd.

Project: Example Subdivision Geotechnical Investigation

Client: Example Developments Ltd.

Location: NE 10-52-25 W4M, near Edmonton, AB

BOREHOLE BH-EX1

XTES #: EX-2026-001

ISSUED · REV 0 · 2026-03-20

Contractor / rig: Example Drilling Ltd. / Track-mounted auger

Coordinates: 334120 E, 5943410 N (UTM NAD83(CSRS) Zone 12)

Method: Solid-stem auger, 150 mm dia.

Ground elev.: 672.40 m Geodetic (CGVD2013)

Date: 2026-03-09 to 2026-03-09

Logged / reviewed: A.B. / C.D., P.Eng.

Depth (m)	Elev. (m)	Symbol	Stand-pipe	SOIL DESCRIPTION (CFEM / Modified USCS)	Sample	Blows/ 150 mm	N	SPT N blows/0.3 m (0 25 50)	PL • w LL (%) (0 30 60)	PP kPa	Su kPa	SO4 %	GW
0	672.40			**TOPSOIL …**									
				CLAY (CI) silty, trace sand, medium plastic, stiff, moist, brown, trace oxide staining (LACUSTRINE)	G1							0.12	
1	671.40				SS2	3-4-5	9			125			
2	670.40				G3								
3	669.40				SS4	4-6-8	14			160			
				CLAY (CI) silty, some sand, trace gravel, medium plastic, very stiff, moist, grey-brown, trace coal and oxide staining, trace gypsum crystals (TILL)									▼
4	668.40				SS5	6-9-12	21			250	110	0.35	
5	667.40												
6	666.40				SS6	8-11-14	25			300			
7	665.40												
				SAND (SM) fine grained, compact, wet, grey, sand lenses (TILL)	SS7	7-10-12	22						▽
8	664.40												S
					SS8	9-12-15	27						
9	663.40			**END OF BOREHOLE AT 9.00 M – TARGET DEPTH REACHED**									

Groundwater: Upon completion: slough at 8.60 m, water at 8.20 m. Seepage while drilling 7.60 m (2026-03-09); latest 4.00 m (2026-10-12).

Installation: Standpipe piezometer, 25 mm PVC Sch 40, slotted 5.50–8.50 m, 0.25 mm (#10) slot, stick-up 0.90 m.

Drilling: 0.00 m Started drilling; 7.60 m Water inflow; 9.00 m Completed borehole.

Notes: Refer to the geotechnical report for interpretation.

Disclaimer: Subsurface conditions are shown at the borehole location only and may vary between boreholes. Read this log together with the accompanying report.

Legend: SS split spoon, ST Shelby tube, G grab, A auger, C core. ▽ seepage while drilling, ▼ water level, S slough. N in blows/0.3 m (field, uncorrected). PP pocket penetrometer, Su field vane, SO4 water-soluble sulphate. Boundaries approximate.

Logged by	Reviewed by	Issue
A.B.	**C.D., P.Eng.**	**Rev 0 – Issued for use – 2026-03-20**

XTES Environmental Ltd.

Project: Example Subdivision Geotechnical Investigation

Client: Example Developments Ltd.

Location: NE 10-52-25 W4M, near Edmonton, AB

BOREHOLE BH-EX2

XTES #: EX-2026-001

Sheet 1 of 1

FOR REVIEW · NOT FOR CONSTRUCTION

Contractor / rig: Example Drilling Ltd. / Track-mounted auger	Method: Solid-stem auger, 150 mm dia.	Date: 2026-03-10 to 2026-03-10
Coordinates: 334190 E, 5943450 N (UTM NAD83(CSRS) Zone 12)	Ground elev.: 671.60 m Geodetic (CGVD2013)	Logged / reviewed: A.B. / C.D., P.Eng.

Depth (m)	Elev. (m)	Symbol	SOIL DESCRIPTION (CFEM / Modified USCS)	Sample	Blows/ 150 mm	N	SPT N blows/0.3 m (0 25 50)	PL • w LL (%) (0 40 80)	PP kPa	Su kPa	SO4 %	GW
0	671.60		**TOPSOIL** silty, moist, black, organics									
			FILL clay, silty, some sand, moist, brown, trace gravel and debris									
1	670.60			SS1	2-3-3	6						
			CLAY (CH) silty, high plastic, firm, moist, dark grey, slickensided (LACUSTRINE)									
2	669.60			ST2						40	0.48	
3	668.60			SS3	2-4-4	8			90			
4	667.60		**CLAY (CI)** sandy, some gravel, medium plastic, stiff, moist, grey, trace coal (TILL)									
				SS4	5-7-9	16			180			
5	666.60											
6	665.60			SS5	6-9-11	20			220			▽
7	664.60											▼
				SS6	8-10-13	23						
			END OF BOREHOLE AT 7.50 M – TARGET DEPTH REACHED									
8	663.60											
9	662.60											

DRAFT

Groundwater: Upon completion: water at 6.80 m. Seepage while drilling 6.10 m (2026-03-10).

Notes: Refer to the geotechnical report for interpretation.

Disclaimer: Subsurface conditions are shown at the borehole location only and may vary between boreholes. Read this log together with the accompanying report.

Legend: SS split spoon, ST Shelby tube, G grab, A auger, C core. ▽ seepage while drilling, ▼ water level, S slough. N in blows/0.3 m (field, uncorrected). PP pocket penetrometer, Su field vane, SO4 water-soluble sulphate. Boundaries approximate.

Logged by	Reviewed by	Issue
A.B.	**C.D., P.Eng.**	**Draft – not issued**

XTES Environmental Ltd.

Project: Example Subdivision Geotechnical Investigation

Client: Example Developments Ltd.

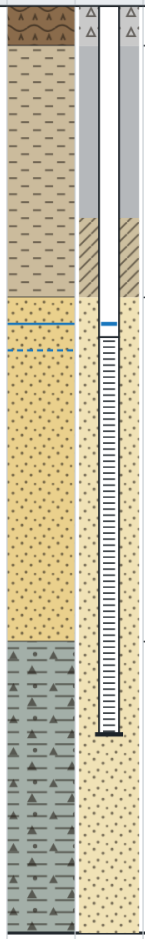
Location: NE 10-52-25 W4M, near Edmonton, AB

BOREHOLE BH-EX3

XTES #: EX-2026-001

DRAFT · NOT FOR CONSTRUCTION

Contractor / rig: Example Drilling Ltd. / Track-mounted auger	Method: Solid-stem auger, 150 mm dia.	Date: 2026-03-10 to 2026-03-10
Coordinates: 334260 E, 5943490 N (UTM NAD83(CSRS) Zone 12)	Ground elev.: 670.80 m Geodetic (CGVD2013)	Logged / reviewed: A.B. / C.D., P.Eng.



Depth (m)	Elev. (m)	SOIL DESCRIPTION (CFEM / Modified USCS)	Sample	Blows/ 150 mm	N	SPT N blows/0.3 m (0 25 50)	PL • w LL (%) (0 30 60)	PP kPa	Su kPa	SO4 %	GW
0	670.80	**TOPSOIL …**									
		SILT (ML) sandy, some clay, low plastic, firm, moist, light brown, trace oxide staining (AEOLIAN)									
1	669.80		SS1	2-3-4	7			75			
2	668.80										
		SAND (SP) fine to medium grained, trace silt, poorly graded, compact, wet, brown (GLACIOFLUVIAL)	SS2	5-7-9	16						▼ ▽
3	667.80										
4	666.80		SS3	8-11-13	24						
5	665.80	**CLAY (CI)** silty, some sand, trace gravel, medium plastic, stiff, moist, grey, trace coal (TILL)									S
			SS4	6-8-10	18			175			
6	664.80										
			SS5	7-9-12	21						
7	663.80	**END OF BOREHOLE AT 7.00 M – TARGET DEPTH REACHED**									
8	662.80										
9	661.80										

DRAFT

Groundwater: Upon completion: slough at 5.20 m, water at 2.90 m. Seepage while drilling 2.60 m (2026-03-10); latest 2.40 m (2026-10-12).

Installation: Standpipe piezometer, 25 mm PVC Sch 40, slotted 2.50–5.50 m, 0.25 mm (#10) slot, stick-up 0.90 m.

Drilling: 2.60 m Water inflow, hole caving.

Notes: Refer to the geotechnical report for interpretation.

Disclaimer: Subsurface conditions are shown at the borehole location only and may vary between boreholes. Read this log together with the accompanying report.

Legend: SS split spoon, ST Shelby tube, G grab, A auger, C core. ▽ seepage while drilling, ▼ water level, S slough. N in blows/0.3 m (field, uncorrected). PP pocket penetrometer, Su field vane, SO4 water-soluble sulphate. Boundaries approximate.

Logged by	Reviewed by	Issue
A.B.	**C.D., P.Eng.**	**Draft – not issued**

Borehole	Sample	Depth (m)	Logged	N	N60	(N1)60	w %	LL	PL	PI	Class	G/S/F %	SO4 %	CSA	Swell	Frost
BH-EX1	G1	0.75	CI	—	—	—	24.6	—	—	—	—	—	0.12	S-3	—	F3–F4*
BH-EX1	SS2	1.50	CI	9	10	17	26.1	41	18	23	CI	—	—	—	Medium	F3
BH-EX1	G3	2.25	CI	—	—	—	25.3	—	—	—	—	—	—	—	—	F3–F4*
BH-EX1	SS4	3.00	CI	14	17	22	22	—	—	—	—	—	—	—	—	F3–F4*
BH-EX1	SS5	4.50	CI	21	28	32	16.4	36	14	22	CI	—	0.35	S-2	Medium	F3
BH-EX1	SS6	6.00	CI	25	33	35	15.2	—	—	—	—	—	—	—	—	F3–F4*
BH-EX1	SS7	7.60	SM	22	29	29	18.9	—	—	—	—	16/58/26	—	—	—	F2
BH-EX1	SS8	8.80	SM	27	38	35	17.5	—	—	—	—	—	—	—	—	F2–F3*
BH-EX2	SS1	1.50	FILL	6	7	11	21	—	—	—	—	—	—	—	—	—
BH-EX2	ST2	2.25	CH	—	—	—	34.2	62	22	40	CH	—	0.48	S-2	High	F3
BH-EX2	SS3	3.00	CH	8	10	13	32.8	—	—	—	—	—	—	—	—	F3–F4*
BH-EX2	SS4	4.50	CI	16	21	23	18.3	37	15	22	CI	—	—	—	Medium	F3
BH-EX2	SS5	6.00	CI	20	27	25	16.9	—	—	—	—	—	—	—	—	F3–F4*
BH-EX2	SS6	7.40	CI	23	31	26	16	—	—	—	—	—	—	—	—	F3–F4*
BH-EX3	SS1	1.00	ML	7	7	12	18.2	—	—	—	—	—	—	—	—	F4
BH-EX3	SS2	2.50	SP	16	19	30	21.4	—	—	—	—	5/89/6	—	—	—	NFS–F2*
BH-EX3	SS3	4.00	SP	24	29	38	19	—	—	—	—	—	—	—	—	NFS–F2*
BH-EX3	SS4	5.50	CI	18	24	29	17.1	35	15	20	CI	—	—	—	Medium	F3
BH-EX3	SS5	6.80	CI	21	28	31	16.6	—	—	—	—	—	—	—	—	F3–F4*

N60 = N·(ER/60)·CB·CS·CR (Youd et al. 2001); (N1)60 = N60·CN, CN = √(100/σ'v) ≤ 1.7. Class: Modified USCS from LL and PI. CSA: exposure class per CSA A23.1 Table 3 (S-1 ≥ 2.0 %, S-2 0.20–2.0 %, S-3 0.10–0.20 %). Swell (after Chen 1975) and frost group (USACE F1–F4) are indicative; * needs grain size or PI to confirm.

Borehole	Date	Reading	Depth (m)	Ground el.	Water el.	Measured from
BH-EX1	2026-03-09	Seepage while drilling	7.60	672.40	664.80	Ground surface
BH-EX1	2026-03-09	Upon completion	8.20	672.40	664.20	Ground surface
BH-EX1	2026-03-24	Standpipe reading	4.10	672.40	668.30	Ground surface
BH-EX1	2026-04-14	Standpipe reading	3.90	672.40	668.50	Ground surface
BH-EX1	2026-05-20	Standpipe reading	3.60	672.40	668.80	Ground surface
BH-EX1	2026-07-15	Standpipe reading	3.80	672.40	668.60	Ground surface
BH-EX1	2026-09-10	Standpipe reading	4.20	672.40	668.20	Ground surface
BH-EX1	2026-10-12	Standpipe reading	4.00	672.40	668.40	Ground surface
BH-EX2	2026-03-10	Seepage while drilling	6.10	671.60	665.50	Ground surface
BH-EX2	2026-03-10	Upon completion	6.80	671.60	664.80	Ground surface
BH-EX3	2026-03-10	Seepage while drilling	2.60	670.80	668.20	Ground surface
BH-EX3	2026-03-10	Upon completion	2.90	670.80	667.90	Ground surface
BH-EX3	2026-03-24	Standpipe reading	2.40	670.80	668.40	Ground surface
BH-EX3	2026-04-14	Standpipe reading	2.10	670.80	668.70	Ground surface
BH-EX3	2026-05-20	Standpipe reading	1.90	670.80	668.90	Ground surface
BH-EX3	2026-07-15	Standpipe reading	2.20	670.80	668.60	Ground surface
BH-EX3	2026-09-10	Standpipe reading	2.60	670.80	668.20	Ground surface
BH-EX3	2026-10-12	Standpipe reading	2.40	670.80	668.40	Ground surface

Depths below ground surface. Seepage while drilling is not an equilibrium level.